

Quantum Computing Assignment

Introduction

This assignment is based on the BB84 quantum key distribution protocol, which we covered in Lecture 3. The aim of the assignment is to implement a simulation of BB84, demonstrating correct operation of the protocol and showing that an attacker can be detected. You can work on the assignment during your lab sessions on Friday 8th May (after working through the Grover's algorithm notebook, which won't take long) and Thursday 14th May.

The task

Implement a demonstration of the BB84 protocol, with and without an attacker. Although BB84 is a communication protocol involving two separate agents (or three, including an attacker), implement it as a single sequential program — but make it clear which parts of the code correspond to each agent. To generate the random choices that are needed by the protocol, measure a suitable quantum state such as $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Don't use a standard Python random number generator. The attacker should try to obtain information about the key, somehow, for example by measuring some of the qubits. You can also try other kinds of attack. The participants in the protocol should report an attack by using some threshold on the proportion of disrupted qubits.

Practical instructions

The GitHub repository `SimonGay/SIT-UofG-QC-Assignment` is a template. You can create your own repository as a copy of it. It contains two notebooks, `BB84-Plain` and `BB84-Attacker`. They are empty except for the usual imports. Work on the assignment in your repository, then submit it by giving Simon Gay access (user name: SimonGay). The submission deadline is **Thursday 14th May** at 18.00. The last commit to your repository must be logged before that time.

Remember to fill in this form:

`https://forms.cloud.microsoft/e/9k5Lp8UqJ0`

with your name, student numbers and GitHub user name.